NQ/LAB / PUBLICATION NOTE / 8 NOVEMBER 2026

Post-event price-path research update

The following 18 pages preserve the approved November 7 presentation. The selected aggregate findings are now released with methods, limitations, verification records, four summary CSVs, and an aggregate first-crossing figure.

This note supersedes the historical "unpublished" label on slide 12 and the initial 15-file / future-release description on supplement 4.

Fresh internal reproduction: 27 synthetic tests passed. All 14 numerical, table, and figure artifacts matched. The separate arithmetic checker passed 69,900 comparisons on shared stored inputs. These checks do not establish independent signal validation or a complete scientific review of Experiment D.

Historical provenance and independent Pine runtime parity remain unresolved. No non-event benchmark or predictive advantage is established. Underlying prices, event records, detector source, and private reports remain private.

https://github.com/Anandaravinderusf/nqlab-market-structure-research

Algorithmic Market-Structure Detection and Short-Term Price Behavior in Nasdaq-100 Futures

From Chart Patterns to Measured Evidence

RESEARCH QUESTION

Do directional price responses persist across different historical samples?

Ananda Ravinder

https://github.com/Anandaravinderusf/nqlab-market-structure-research

Why I moved beyond individual chart examples

Studying and trading Nasdaq-100 futures made me curious about the patterns I saw.

A convincing example

A pattern looks meaningful on a historical chart.

→

01 **Define events consistently**

02 **Measure every included outcome**

03 **Compare selected samples**

Conceptual research workflow. A selected example does not establish a consistent relationship.

The research question

Does price keep moving in the event’s direction?

5 / 15

minutes after the event close

Does that response persist across selected historical samples?

Rule-defined displacement and R events. Descriptive signed changes, without a trading-performance or causal claim. [1,5]

What a one-minute candle records

Conceptual upward candle. No historical prices or detected signals.

HIGH

Highest price

NQ futures: contracts linked to the Nasdaq-100 stock index.



CLOSE

Last price

Body
Open to close

OPEN

First price

LOW

Lowest price

VOLUME Trading activity in the minute, recorded separately from price.

Concept: CME Group, Chart Types.

Intended market: NQ. Historical mapping unverified. [1,5]

One rule set, two event families

Conceptual sequence, not an observed event or a complete detector specification.

Available structure

A previously available structural high or low.

→ if later conditions qualify

Displacement

An accepted close beyond that level.

→

Later R

A qualifying confirmation of the pending setup.

Both directions of both event types enter the cohorts.

Bullish = upward. Bearish = downward. Each accepted event uses its own completed close.

R depends on a prior setup; it is not automatic after every displacement. The reported means combine all four event types. [1–3]

One authentic example, with an explicit clock

January 2, 2025. All times ET.



Bearish displacement

10:17 open
10:18 available at close

Bearish R confirmation

10:19 open
10:20 available at close

Each is a separate observation, measured from its own close.

Selected illustration, not a typical outcome. Original candles/markers preserved; no structural level reconstructed. Historical availability is modeled. [1,4]

Historical inputs to measured outcomes

01 INPUT

One-minute OHLCV

Warm-up history and coverage checks

02 DETECT

Frozen SDR v0.11

Accepted displacement and R events

03 RECORD

Timestamped events

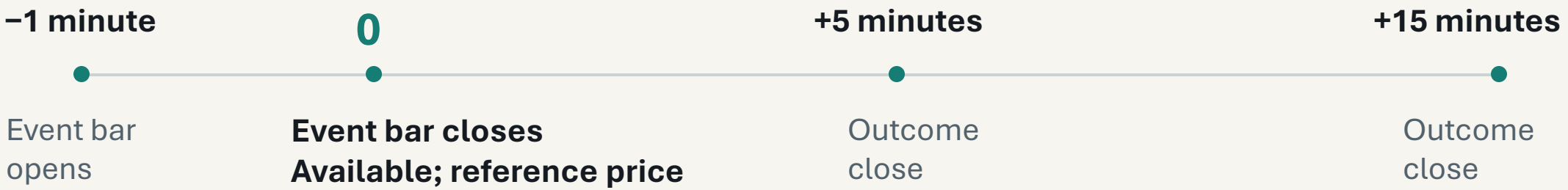
Type, direction, bar open and modeled availability

04 MEASURE

Signed changes

5- and 15-minute forward closes

Signed change = direction × (later close − event-bar close)



Conceptual clock, not to scale. Each displacement or R uses its own close. Direction: +1 bullish, −1 bearish. No assumed executable fill or P&L. [1]

January pilot: the first empirical checkpoint

87 events

8 selected mornings

Selected for available coverage.
All four event types.

5-MINUTE RAW MEAN

−0.11

15-MINUTE RAW MEAN

+2.15

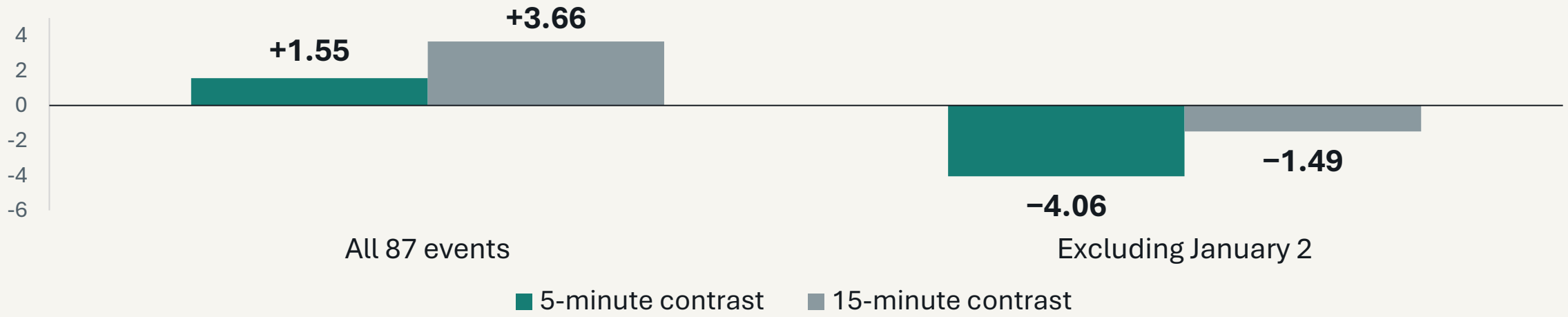
A positive 15-minute sample mean, with no baseline adjustment.

Event-weighted signed changes, provisional NQ-labeled points. An exploratory sample, without an established predictive advantage. Exact values in S1. [2,3]

Experiment C: what is the baseline comparison?

Same New York reference-close minute and the event's direction

Event signed outcome − **Average of seven other-day controls** = **Event-minus-baseline contrast**



A separate estimand from raw forward changes. Provisional NQ-labeled points.

Joint Jan 2 origin/donor omission: n=80, six donors, controls recalculated. Reused retrospective controls may contain events and later dates. [1,3]

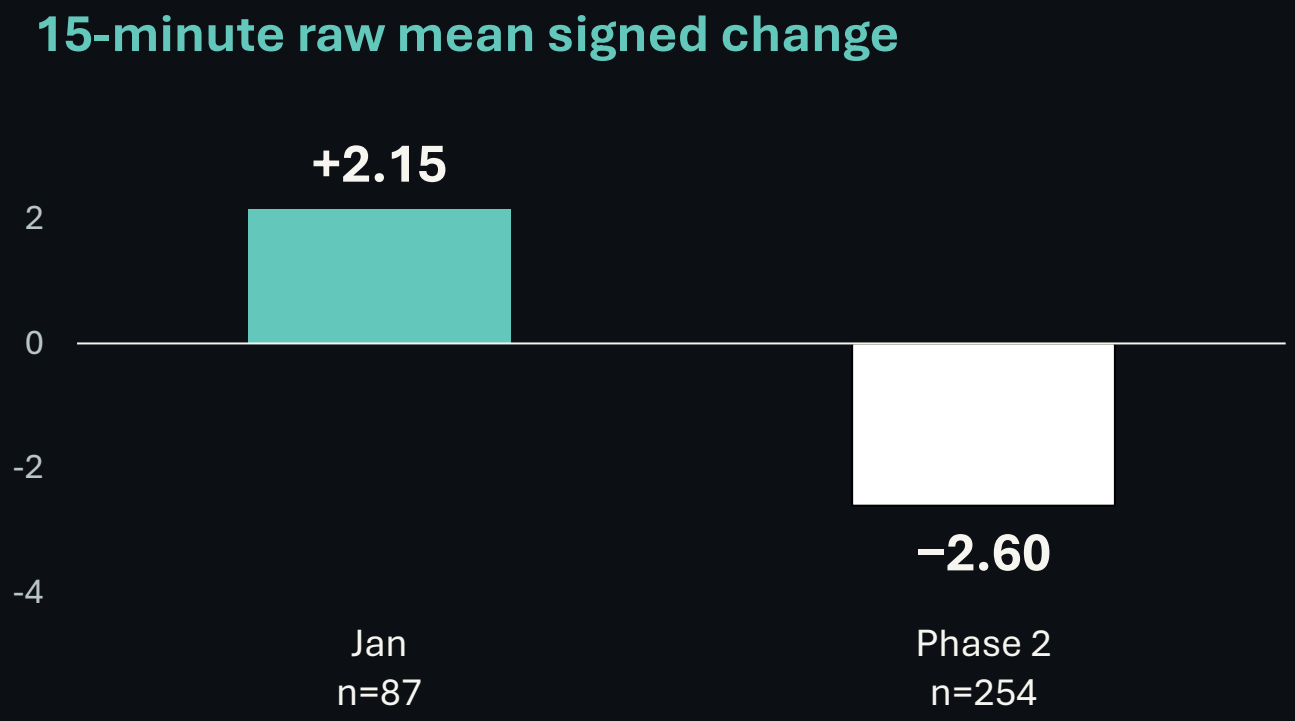
Phase 2: the pilot’s positive mean did not recur

254 events

22 / 25 sessions retained

Calendar-selected across 2021–2025.
3 data-quality exclusions,
without replacement.

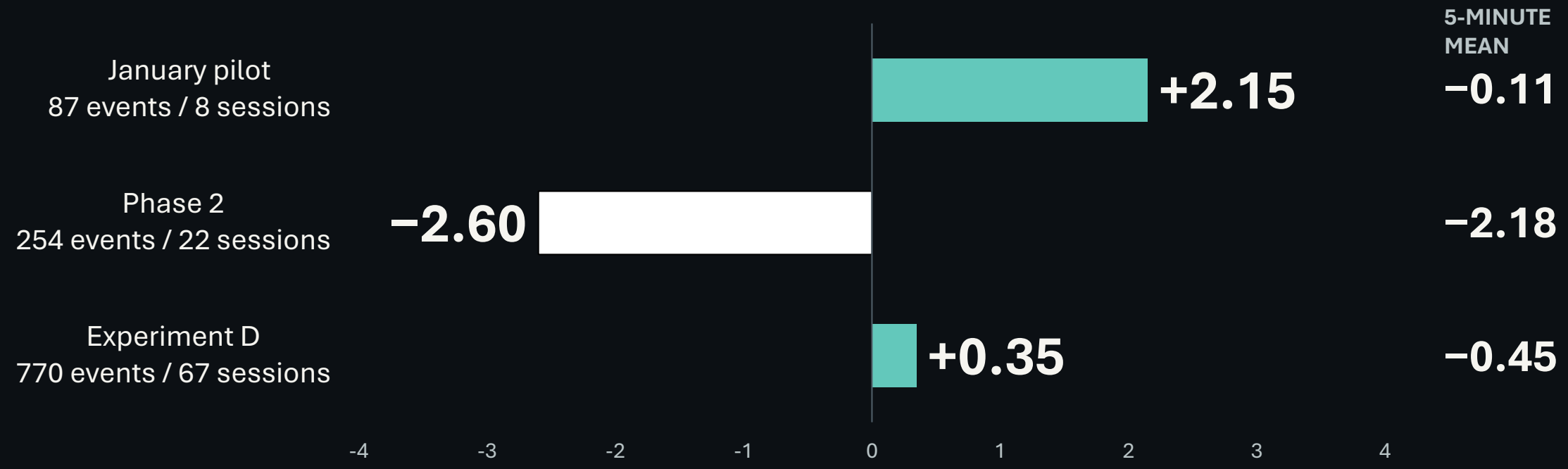
5-minute raw mean: −2.18



Provisional NQ-labeled points. Selected retrospective sessions, not prospective validation. Local timestamps/hashes do not independently establish preregistration. [1–3]

The pilot mean did not recur consistently

15-minute mean signed change, provisional NQ-labeled points



Separate raw means; no combined effect across cohorts.

Event-weighted raw means, selected dependent observations. D: 67/80 sessions, 13 exclusions without replacement; dates disjoint.

Experiment D: completed for independent review, not independently reviewed.

Aggregate CSV

Verification

Small endpoint mean, substantial movement

770 events / 67 selected sessions. 15-minute horizon. Provisional NQ-labeled points.

AT THE 15-MINUTE CLOSE

+0.35

Mean signed endpoint change

Endpoint SD 43.26

Dispersion across endpoint outcomes

WITHIN THE 15-MINUTE WINDOW

Median maximum favorable excursion **23.50**

Median maximum adverse excursion **23.375**

With / against the event's direction

51.4%

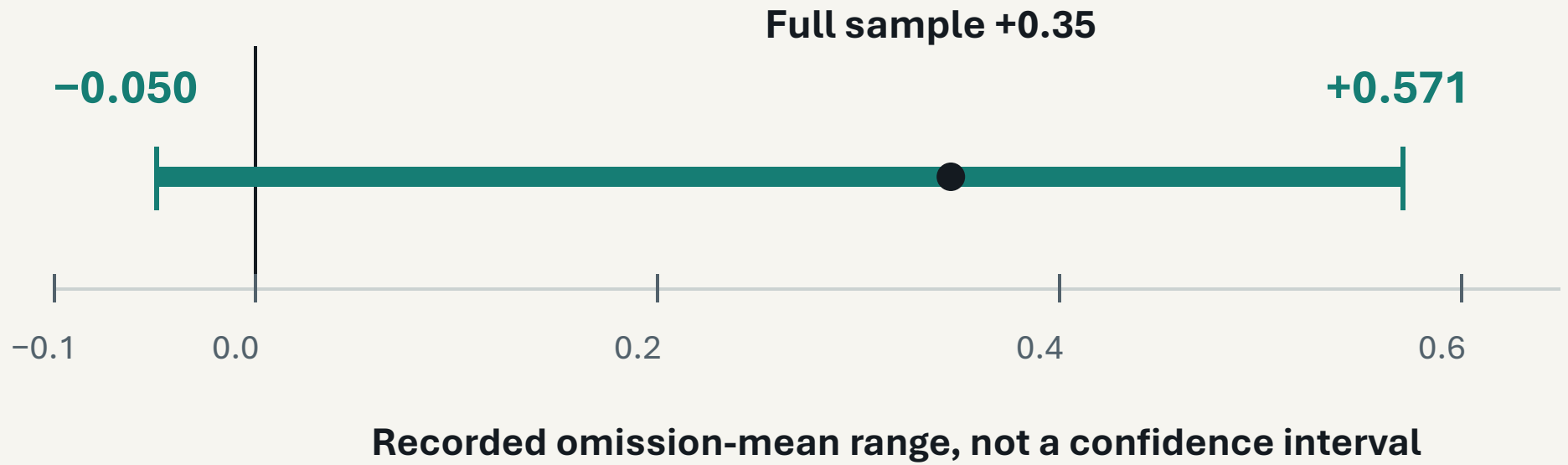
396 / 770 events reached both opposing 10-point levels within 15 minutes.

Next 15 candle highs/lows; event candle excluded. Separate marginal medians. OHLC does not resolve within-candle order or fills.

Source: NQ/LAB post-event-path v1, 7 Nov 2026 (unpublished). No non-event benchmark or independent signal validation.

Omitting one session can change the sign

15-minute raw means after one-session deletions



Recorded omission-mean range, not a confidence interval

760 / 770

origins had another same-day origin
less than 15 minutes away.

15m endpoints: −0.049541 / +0.570623 points. Most events remain shared across deletions. Five-minute omission means remained negative. [3,5]

What this research contributes

BUILT

A documented historical event-study framework

Explicit event timing and signed forward measurement.

OBSERVED

The positive 15-minute pilot mean did not recur consistently

Sample-sensitive means, overlapping windows and two-way price paths.

NEXT

Conditional relationships by event type, market conditions or structural context

Future hypotheses requiring stronger validation.

Historical provenance and independent Pine runtime parity remain unresolved. No predictive advantage established. [1–3,5]

https://github.com/Anandaravinderusf/nqlab-market-structure-research

Exact results and separate estimands

Raw mean signed changes, six decimals. Provisional NQ-labeled points.

Cohort	Events	Sessions	5 minutes	15 minutes
January pilot	87	8	−0.109195	+2.149425
Phase 2	254	22	−2.182087	−2.600394
Experiment D	770	67	−0.445130	+0.346753

January Experiment C v2: event-minus-baseline contrasts

All 87 events: +1.553777 / +3.662562 (5m / 15m)

Joint Jan 2 origin/donor omission, n=80: −4.056250 / −1.491667

Controls use other selected January days at the same reference-close minute and direction. Reused controls may contain events and later dates. Distinct event-weighted estimands. [2,3]

Historical replay and genuine live collection

HISTORICAL PYTHON REPLAY

January 2, 2025



◇ Bearish displacement · SDR 0.11

10:17 open / 10:18 available ET

GENUINE LIVE WEBHOOK

November 4, 2026

SOURCE	EVENT
SDR v0.11 Live webhooks	Bullish R confirmation

ALERT TRIGGER / ET	COLLECTOR RECEIPT / ET
Nov 04, 20:39:00	Nov 04, 20:39:00

Bar open 20:38:00 ET

Trigger / receipt 20:39:00 ET

Modeled replay clocks differ from live trigger/receipt. Whole-second display does not prove zero latency or historical runtime parity. [1,4]

Historical inputs and interpretation limits

Jan 2, 2025 · One-minute candles · Historical Python replay · Selected illustration [4]



Study event opens: 09:30–10:59 ET. Chart includes later context. Warm-up history is not shown.

Expansions: all 105 morning/buffer buckets and ≥120 contiguous pre-open minutes. No gap filling. [1]

Unresolved: contract/rollover mapping, original clocks, vendor provenance and independent Pine parity. [3,5]

Public research and evidence scope

https://github.com/Anandaravinderusf/nqlab-market-structure-research

15 curated files: overview, methods, aggregate results, limitations, source and verification records, reproducibility scope and citation metadata.

[1] Methodology

[2] Aggregate results

[3] Verification

[5] Limitations

Reproducibility scope

Research sources

Citation metadata

Reuse disclosure

Open the repository

Illustrations [4]: Jan 2, 2025 historical Python replay; Nov 4, 2026 genuine live collection. Owner-supplied assets.

Original reports, candles, event records and full detector code remain private. Current checks verify preserved arithmetic without regenerating signals. Full external replication and independent signal-generation validation remain unavailable.

Initial public commit: 6f09433d2762 (6 Nov 2026). Future reviewed releases are planned.